

CTCF binding architecture of clustered TF-paralog loci in resting CD4⁺ T cells: ohnolog vs. tandem (SSD) clusters

Iteration 19 — Paralogous TF project

ETS P01 genomics platform

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Abstract

This report repeats the iteration-19 CTCF analysis in a second, immune-relevant cell type — primary **resting CD4⁺ $\alpha\beta$ T cells** — and asks whether the conclusions drawn from the GM12878 baseline reproduce. Using an ENCODE CD4 T-cell CTCF ChIP-seq peak set (hg38) over the same clustered-locus windows, we find the same picture: both **clustered cis-ohnolog** and **clustered SSD-tandem** loci are CTCF-enriched over the genome-wide expectation ($\sim 1.5\text{--}2.0\times$ obs/exp) and are *globally indistinguishable* in CTCF density (Mann–Whitney U, all $p > 0.15$; 0/60 metagene bins at BH-FDR < 0.05). The one separating feature is again the *number of CTCF peaks between the two paralog members* (ohnolog median 3 vs. SSD-tandem 0; $p = 0.0089$). In CD4 T cells the contrast is sharper than in GM12878: SSD-tandem pairs frequently carry *no* CTCF peak between their promoters, so in the 2-member-pair subset the difference persists even after distance normalization ($p = 0.021$). As in GM12878, the effect is entangled with the wider inter-member spacing of ohnolog pairs. The core result — ohnolog paralogs are separated by more insulator elements than tandem duplicates — **replicates across both cell types**.

1 Data and provenance

CTCF peaks. ENCODE experiment ENCSCR470KCE (TF ChIP-seq, target CTCF, biosample *Homo sapiens* CD4-positive $\alpha\beta$ T cell, male adult 20 yr — primary *resting/unstimulated* CD4 T cell), file ENCFF858TLX (*IDR-thresholded peaks*, **bed narrowPeak**, GRCh38, released). Downloaded 2026-07-18. Sanity checks: 40,346 peaks; chr1–22, chrX, chrY (male donor). *Note:* ENCODE has no experiment explicitly labelled “naive CD4”; this resting primary CD4 T cell is the closest match (its activated counterpart, ENCSCR806WWX, is separate). Genome-wide expected CTCF rate for this set: **0.0131 peaks/kb**.

Locus set. Identical to the GM12878 analysis: the iteration-16 clustered-locus catalog (hg38), two clustered classes only, extended ± 100 kb. For direct cross-cell-type comparability the single chrY locus (HSFY1/HSFY2) remains excluded even though this male sample has chrY signal. Final set: **47 clustered cis-ohnolog** and **30 clustered SSD-tandem** clusters. Methods (windows, fixed + metagene binning, per-locus metrics, tests) are exactly as in the GM12878 report; only the CTCF peak file changed.

2 Results

Global CTCF density does not separate the two clustered classes

Over the ± 100 kb window both groups sit well above the genome-wide expectation (median obs/exp 1.96 ohnolog, 2.05 SSD-tandem) and none of the global density metrics differ (Table 1; all $p > 0.15$). The metagene profiles overlap within their bootstrap CI bands (Fig. 1) and **0 of 60** bins reach BH-FDR < 0.05 . Per-flank enrichment is likewise indistinguishable between groups and shows no left/right asymmetry within either group (LEFT $p = 0.97$, RIGHT $p = 0.81$; within-group Wilcoxon $p > 0.85$).

Table 1: Per-locus CTCF metrics in resting CD4 T cells, **clustered cis-ohnolog** ($n = 47$) vs. **clustered SSD-tandem** ($n = 30$). Two-sided Mann–Whitney U; r_{rb} = rank-biserial effect size (negative $\Rightarrow$ ohnolog $>$ SSD-tandem).

Metric	median (ohnolog)	median (SSD)	U	p	r_{rb}
CTCF peaks/kb, window	0.0256	0.0268	718.5	0.892	−0.019
CTCF peaks/kb, locus body	0.0251	0.0105	839.0	0.159	−0.190
CTCF peaks/kb, flanks	0.0250	0.0250	666.0	0.686	+0.055
Observed/expected, window	1.962	2.052	719.0	0.888	−0.020
Nearest CTCF to a TSS (bp, min)	9445	10478	707.0	0.988	−0.003
Nearest CTCF to TSS (bp, mean)	22427	22616	719.0	0.888	−0.020
CTCF peaks between members	3.0	0.0	948.5	0.0089	−0.345
Inter-member TSS distance (bp)	74355	49409	866.0	0.094	−0.228
CTCF peaks/kb between members (norm.)	0.0190	0.0000	876.0	0.067	−0.243

Ohnolog members are separated by more CTCF — and SSD pairs often by none

As in GM12878, the distinguishing metric is the count of CTCF peaks between the two paralog members (ohnolog median 3 vs. SSD-tandem 0; $p = 0.0089$, $r_{rb} = -0.35$; Fig. 2). The CD4 contrast is stronger at the promoter level: restricting to 2-member pairs, ohnolog pairs have a median of 2 CTCF peaks between their promoters versus 0 for SSD-tandem pairs ($p = 0.010$), and here the per-kb (distance-normalized) difference is also significant (0.023 vs. 0.0 peaks/kb, $p = 0.021$; Fig. 3). Ohnolog promoters are again $\sim 2\times$ farther apart (median 52.6 kb vs. 25.2 kb, $p = 0.15$), so part of the raw-count gap remains a spacing effect; but the many SSD-tandem pairs with literally zero intervening CTCF indicate that recent tandem duplicates tend to share a single uninsulated regulatory neighbourhood, whereas ohnolog paralogs are more often split by an insulator.

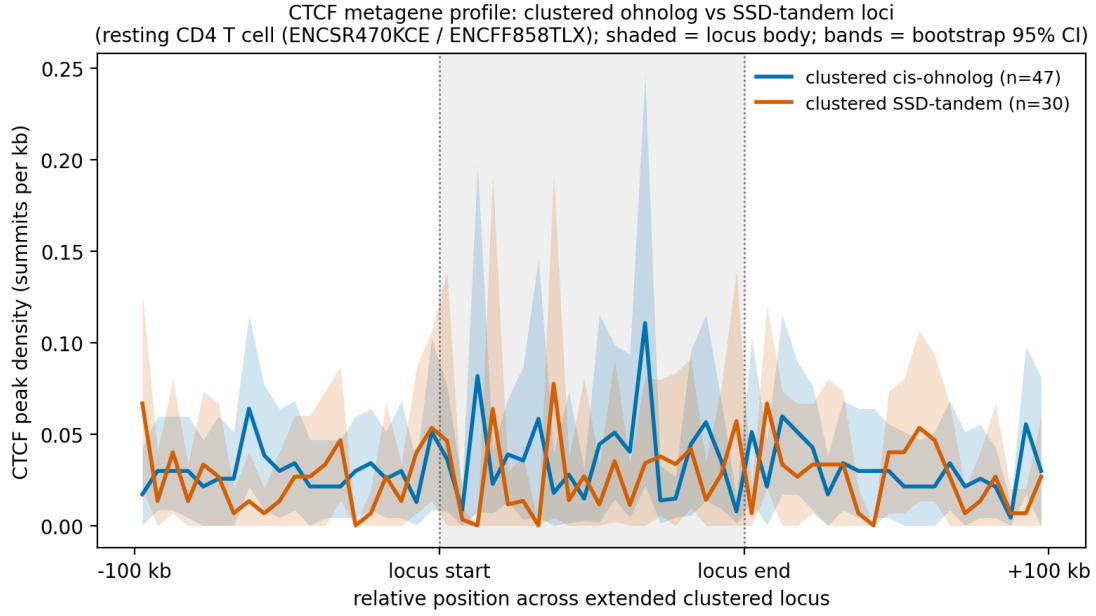


Figure 1: CD4 T-cell CTCF metagene profile (length-normalized). Lines = mean CTCF summit density/kb; bands = bootstrap 95% CI; grey = locus body. The two clustered classes overlap throughout; no metagene bin survives BH-FDR < 0.05.

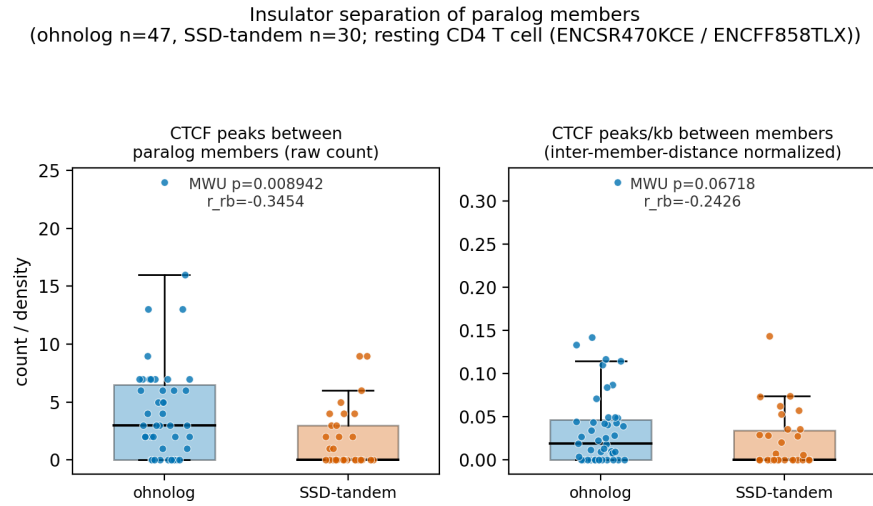


Figure 2: CTCF peaks between paralog members in CD4 T cells. Left: raw count differs ($p = 0.0089$). Right: distance-normalized per-kb density (all clustered loci) is borderline ($p = 0.067$); in the 2-member-pair subset it reaches significance (Fig. 3).

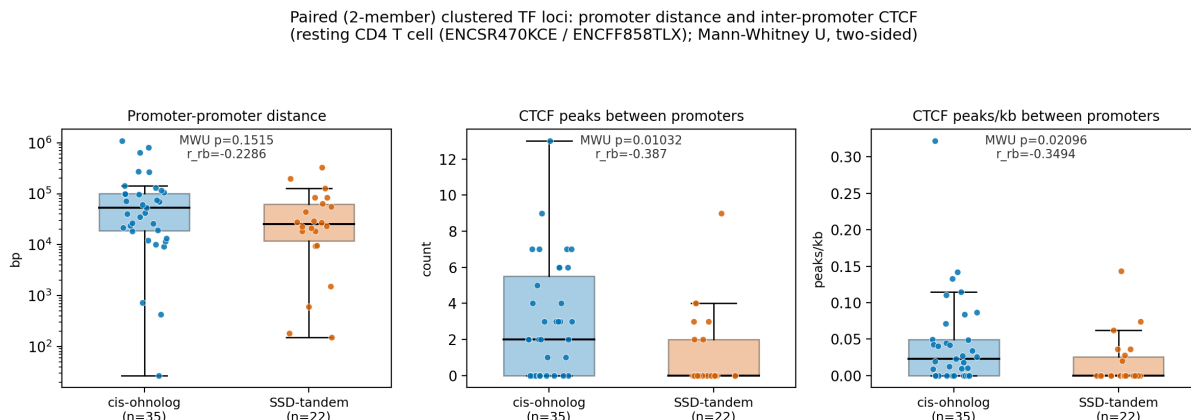


Figure 3: Paired (2-member) clustered loci in CD4 T cells: promoter-promoter distance and CTCF between promoters. SSD-tandem pairs frequently have zero CTCF between promoters, so both the raw count ($p = 0.010$) and the distance-normalized density ($p = 0.021$) differ.

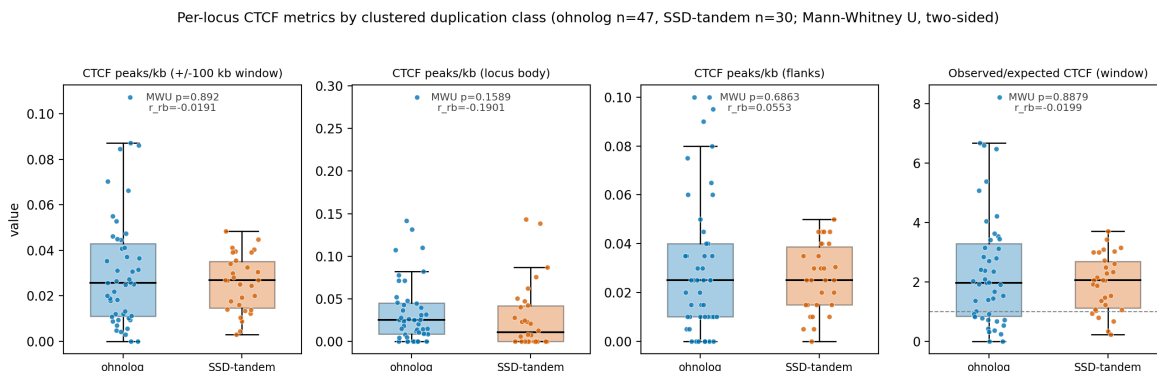


Figure 4: Per-locus global CTCF metrics by class in CD4 T cells; all indistinguishable between groups, both above the genome-wide expectation.

3 Conclusion — cross-cell-type reproducibility

The resting CD4⁺ T-cell CTCF analysis reproduces every qualitative conclusion of the GM12878 baseline: (i) both clustered duplication classes occupy CTCF-enriched, statistically similar overall insulator environments; (ii) the only separating signal is more CTCF peaks between ohnolog than SSD-tandem paralog members; and (iii) that signal is partly a consequence of the wider spacing of ohnolog pairs. The one cell-type difference is a matter of degree, not direction: in CD4 T cells SSD-tandem pairs so often lack any intervening CTCF that the between-promoter contrast survives distance normalization in the pair subset, whereas in GM12878 it did not. That the finding holds in two independent cell types — an EBV B-lymphoblastoid line and primary resting CD4 T cells — argues it reflects a constitutive, mechanism-linked feature of the loci rather than a cell-line artefact.

Caveats. (i) Two cell types now, but still a limited panel; CTCF is largely constitutive. (ii) Small groups (47 vs. 30 clusters) — per-locus tests are underpowered and the inter-member-distance confounder ($p = 0.09$) is not resolved. (iii) Fixed-bp and metagene binnings are reported separately. (iv) The chrY locus was excluded for cross-cell-type comparability; “LEFT/RIGHT” flanks are

genomic, not strand-oriented. (v) ENCODE has no explicitly “naive” CD4 CTCF experiment — the resting primary CD4 T cell (ENCSR470KCE) is used as the closest available proxy.